

Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **The gain budget of the course array.** The ADALM-PHASER carries $N = 8$ microstrip patch elements spaced $d = 14$ mm apart. At the workshop frequency of 10.3 GHz the wavelength is 29.1 mm. Model each element as an ideal element that captures its share of the aperture, so its *power* pattern follows the projected-aperture rule $G_e(\theta) = G_e(0)\cos(\theta)$ with a broadside element gain of 6.0 dBi.

- (a) In one or two sentences, state the difference between the array factor and the antenna pattern, and say what the element factor contributes to the product.

- (b) Compute the broadside peak gain of the array from the element gain and the array gain.

$$G(0^\circ) = \boxed{}$$

- (c) The beam is steered to $\theta_0 = 45^\circ$. Compute the element factor penalty in decibels at that angle.

$$\Delta G = \boxed{}$$

- (d) Compute the peak gain of the array steered to 45° .

$$G(45^\circ) = \boxed{}$$

2. **Scan loss and the practical scan limit.** Use the same array and the same ideal element. Recall from L20 that the array factor's half-power beamwidth off broadside is $\theta_{\text{HP}} \approx 0.886\lambda/(Nd\cos(\theta_0))$, which is 13.2° at broadside for this array.

(a) Compute the element factor penalty at $\theta_0 = 30^\circ$.

$\Delta G =$

(b) Compute the element factor penalty at $\theta_0 = 60^\circ$.

$\Delta G =$

(c) Compute the array factor half-power beamwidth at $\theta_0 = 60^\circ$.

$\theta_{\text{HP}} =$

- (d) A radar engineer specifies this array for coverage over $\pm 75^\circ$. Using your numbers from parts (a) through (c), write two sentences explaining the trade the engineer is accepting, and state why $\pm 60^\circ$ is the usual limit quoted for patch arrays.

3. **Reading a steered sweep.** A student steers the array to $\theta_0 = 30^\circ$ and records the rectangular sweep. The peak reads -0.6 dB relative to the broadside peak and sits at 29.6° . The first sidelobe on the broadside side of the beam reads -12.2 dBc and the first sidelobe on the outboard side reads -15.2 dBc. Uniform array factor theory predicts -12.8 dBc for both.

(a) Explain in one or two sentences why the measured peak sits at 29.6° rather than at the commanded 30° .

(b) Explain why the two first sidelobes differ by 3 dB when the array factor predicts that they are equal.

(c) Express the inboard sidelobe level relative to the array's *broadside* peak rather than relative to the steered peak.

Level =

(d) The same student rotates the array by hand and repeats the measurement. The new trace has the same shape but every feature is shifted 1.8° in the same direction. State whether this is an antenna effect or a measurement effect, and give the reason.

4. **The element and its ground plane.** Take the ideal element power pattern to be $\cos(\theta)$ over the front hemisphere and zero behind the ground plane.

(a) Find the half-power beamwidth of this element.

$$\theta_{\text{HP}} = \boxed{}$$

(b) Compute the directivity of this element, in dBi, integrating over the upper hemisphere only.

$$D_e = \boxed{}$$

(c) A sweep shows a lobe at -20 dBc near $\theta = 82^\circ$. Explain what this feature is, and why it cannot be radiation out the back of the array.

(d) The projected-aperture rule is a model. State one angular range where you would trust a prediction made with it to about a decibel, and one where you would not, and give the reason.

5. **Range, environment, and what a measurement can resolve.** The course array measures $D = 98$ mm between the centers of its outer elements. The lab source is placed 1.0 m from the array face.

- (a) Compute the far-field distance for this array at 10.3 GHz ($\lambda = 29.1$ mm), and state whether the lab range qualifies.

$r =$

- (b) A larger array is to be measured on the same 1.0 m range at 10.3 GHz. Find the largest aperture dimension that still satisfies the far-field criterion.

$D =$

- (c) The GUI sweeps the commanded steer angle in 2.8125° steps. Explain how that step size alone can make a measured half-power beamwidth differ from theory by one to three degrees.

- (d) Name two things you can change about the measurement setup, not the antenna, to make the far-out sidelobes of a trace more trustworthy.

Documentation: _____